

Individual Proposal

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Introduction

What?

This project aims to create an interpretable fair price benchmark for Airbnb listings in Amsterdam. Using the real airbnb dataset, I will estimate what a listing should cost per night based on simple details like property/room type, capacity, bathrooms/bedrooms, amenities, and neighbourhood. Then I will compare the actual price to the expected price and flag listings as underpriced, normal, or overpriced within their neighbourhood.

The goal is to give guests, hosts and travelers a clear, easy to understand reference and highlight areas with mispricing. A simple dashboard will present the results with key charts and maps.

Why?

Amsterdam has many tourists and high demand for short-stay housing. Prices change a lot across neighbourhoods, and many listings are overpriced. Travelers often struggle to find a place at a fair price, and hosts lack a simple benchmark.

By building a price prediction/comparison tool that uses listing details, I can:

- Help guests avoid overpaying and choose fairly priced places
- Help hosts set competitive prices
- Use Amsterdam's historical data to learn realistic price levels across areas
- Provide practical dashboard

This will make pricing choices in Amsterdam faster, fairer, and more data-driven.

Who?

Possible client: Hosts, tourists, travelers.

University supervisors: Project coaches and lecturers

Students: Fontys AI Foundations students

- [Abdurakhmanov, Tymur T.](#)

When?

Start date: 10.09.2025

End date: 6.11.2026

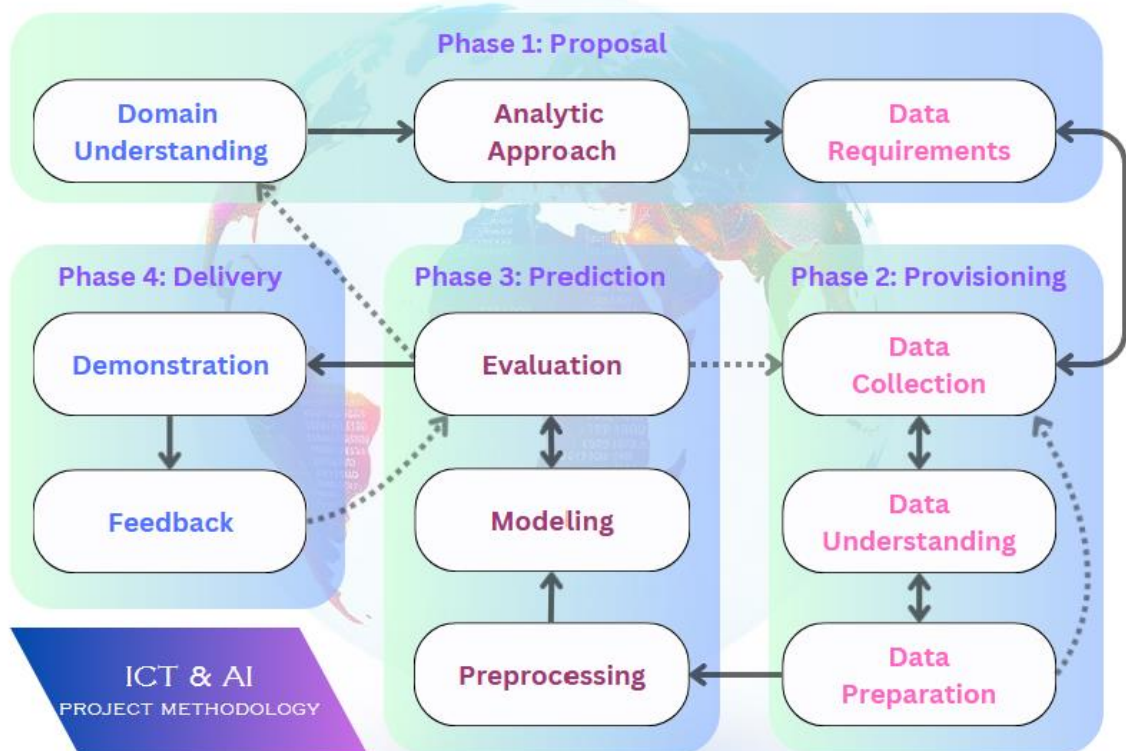
Since this is a student project, dates and deliveries must follow the deadlines published on Canvas. The final delivery aligns with the Portfolio review Final on 25 January 2026.

Key dates (Canvas):

- Portfolio review 1: 2/10/2025
- Portfolio review 2: 6/11/2025

Initial planning assumptions (iterations):

- Iteration 1 :
Initial proposal
- Iteration 2 :
Data cleaning/preparation
Feature engineering
- Iteration 3 :
First models comparison
Set up of chosen model
- Iteration 4 :
Dashboard/explainability



Using Agile methodology, the project will run in short iterations to deliver incremental value, gather regular feedback, and stay flexible. The first iteration begins at project start.

How?

The project will build an interpretable fair-price tool for Airbnb listings in Amsterdam. The process will include:

1. **Collecting and preparing data**
Download Inside Airbnb data and clean key fields (price, bedrooms/bathrooms, amenities, neighbourhood, minimum nights).
2. **Building and training the model**
Use listing features such as property/room type, accommodates, bedrooms/bathrooms, minimum nights, top amenities, and neighbourhood to train a baseline linear model and compare it to a tree models.
3. **Developing a prototype**
Provide a dashboard that shows expected price, the percentage deviation (under / normal / over), and simple charts/maps.

4. **Evaluating performance**

Evaluating performance based on accuracy, usability, and scalability

Domain understanding

Main question

How can I build an AI model that estimates the fair price *for* Airbnb listings in Amsterdam *using listing details and neighbourhood context*?

Methods

- **Stakeholder:** talks with guests/hosts/travelers to understand what “fair price” means for them.
- **Data review & EDA:** assess Inside Airbnb data quality, clean outliers, explore feature–price patterns and neighbourhood differences.
- **Demo:** show early results to collect feedback and keep the tool simple and more useful.

Domain summary

Amsterdam has many tourists and strong demand. Prices vary a lot and many listings are overpriced. People often struggle to find fair offers. Using Inside data from airbnb to build a fair-price reference and showing it in a small dashboard can make choices faster, cheaper, clearer, and more data-driven for both guests and hosts.

Analytic Approach

Target Variable

Primary target is expected nightly price. Predicted fair nightly price for each Amsterdam listing based on listing features. It provides a practical reference for guests and hosts, enable simply see city-level insights and keep it transparent and easy to explain.

Input Features

To make these predictions, the model will use standard listing information from Inside Airbnb, such as property/room type, capacity, bedrooms/bathrooms, amenities, minimum nights, superhost status, and neighbourhood.

If useful, this can be combined later with simple signals from reviews and availability (calendar.csv) and basic geography from neighbourhood shapes to improve context.

Modeling Approach

This is mainly a regression problem, predict the expected nightly price for each Amsterdam listing. I will start with a simple linear model and compare it with a few tree-based (random forest, gradient boosting) and linear (LinearRegression, Ridge, Lasso) models to check robustness.

Evaluation Metrics

To evaluate the predictions, I will measure how close estimated prices are to actual prices.

- For the numeric outcome (nightly price): MAE, RMSE and R^2 .
- Results will be shown in a simple dashboard.

Model Interpretability & Explainability

I will keep the model simple, transparent, and use basic feature importance. If needed, I may add SHAP on a few examples to show which listing details most influence the predicted fair price.

Data Requirements

Source

Inside Airbnb (Amsterdam): provides public datasets for the city. Key files:

- **listings.csv - listings.csv**
id, listing_url, scrape_id, last_scraped, source, name, description, neighborhood_overview, picture_url, host_id, host_url, host_name, host_since, host_location, host_about, host_response_time, host_response_rate, host_acceptance_rate, host_is_superhost, host_thumbnail_url, host_picture_url, host_neighbourhood, host_listings_count, host_total_listings_count, host_verifications, host_has_profile_pic, host_identity_verified, neighbourhood, neighbourhood_cleansed, neighbourhood_group_cleansed, latitude, longitude, property_type, room_type, accommodates, bathrooms, bathrooms_text, bedrooms, beds, amenities, price, minimum_nights, maximum_nights, minimum_minimum_nights, maximum_minimum_nights, minimum_maximum_nights, maximum_maximum_nights, minimum_nights_avg_ntm, maximum_nights_avg_ntm, calendar_updated, has_availability, availability_30, availability_60, availability_90, availability_365, calendar_last_scraped, number_of_reviews, number_of_reviews_ltm, number_of_reviews_l30d, availability_eoy, number_of_reviews_ly, estimated_occupancy_l365d, estimated_revenue_l365d, first_review, last_review, review_scores_rating, review_scores_accuracy, review_scores_cleanliness, review_scores_checkin, review_scores_communication, review_scores_location, review_scores_value, license, instant_bookable, calculated_host_listings_count, calculated_host_listings_count_entire_homes, calculated_host_listings_count_private_rooms, calculated_host_listings_count_shared_rooms, reviews_per_month
- **calendar.csv** - listing_id, date, available, price, adjusted_price, minimum_nights, maximum_nights
- **neighbourhoods.geojson** - neighborhood boundaries for maps.
- **neighbourhoods.csv** - neighborhood names
- **reviews.csv** - review text per listing.

Expected from source

Apartment properties: property type, room type, bedrooms, bathrooms, minimum nights, maximum nights, amenities, max occupancy, availability through year

Target: price through the last year/s

Location: longitude and latitude, neighborhood/area name

Optional: how many times rented per year, reviews

What needed from this source

Extract fields such as price, property/room type, accommodates, bedrooms/bathrooms, amenities, minimum nights, neighborhood, and optionally review stats and calendar dates. These will let estimate an expected nightly price and compare listings within the same neighbourhood.

Data preparation

Before modeling, I will:

- Clean fields.
- Standardize formats and types (CSV, text to numeric, categories)
- Align time periods if I will be using calendar.csv.
- Build a single analysis table.

Final dataset

The prepared dataset will include:

- **Listing** (property/room type, capacity, bedrooms/bathrooms, amenities, minimum/maximum nights, neighborhood).
- **Target** (price).
- **Optional**(review signals, calendar based seasonality features).

This dataset will be used to train, validate, and report the fair-price model for Amsterdam.